A counterexample to the Josefson–Nissenzweig factor converse for $C_p(X, E)$

Candidate full negative answer to arXiv:2107.03211, Problem 4.8

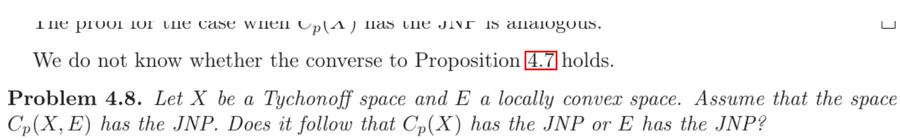
19 July 2026

Status. Counterexample, likely valid, pending human review. The construction gives a negative answer to the problem as stated and, more strongly, produces a complemented copy of $(c_0)_p$. The bounded novelty search below found no exact prior answer, but novelty is not certified.

1 Source problem

Bargetz, Kąkol, and Sobota ask the following in Problem 4.8 on page 7 of arXiv:2107.03211v3 [1].

> Let X be a Tychonoff space and E a locally convex space. Assume that $C_p(X, E)$ has the Josefson–Nissenzweig property. Does it follow that $C_p(X)$ has the Josefson–Nissenzweig property or E has the Josefson–Nissenzweig property?


The proof for the case when $C_p(X)$ has the JNP is analogous. □

We do not know whether the converse to Proposition 4.7 holds.

Problem 4.8. *Let X be a Tychonoff space and E a locally convex space. Assume that the space $C_p(X, E)$ has the JNP. Does it follow that $C_p(X)$ has the JNP or E has the JNP?*

Figure 1: Problem 4.8 and its immediate context in the source paper.

2 Definitions

All spaces may be taken over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. For a locally convex space G, a set $B \subseteq G$ is *barrel-bounded* if every barrel in G absorbs B. The topology $\beta^*(G', G)$ is uniform convergence on the barrel-bounded subsets of G. The space G has the JNP if some sequence in G' is null for $\sigma(G', G)$ but is not null for $\beta^*(G', G)$ [2].

Write $(c_0)_p$ for the Banach space c_0 equipped only with its pointwise topology inherited from $\mathbb{K}^{\mathbb{N}}$. Let c_{00} denote the finitely supported scalar sequences, equipped here with the norm inherited from ℓ_1. Thus c_{00} is a normed locally convex space, but it is not complete. The vector e_n is the nth standard unit vector, and $\pi_n(x) = x_n$.

We use the Stone–Cech compactification $\beta\mathbb{N}$ of the discrete space $\mathbb{N}$. Restriction identifies $C(\beta\mathbb{N})$ isometrically with ℓ_∞.

3 Proof intuition

The earlier finite-slice reduction for this target showed that a counterexample must escape in both tensor variables: it needs infinitely many points of X and infinitely many independent coefficient functionals in E'. The diagonal family

$$f \longmapsto \pi_n(f(n))$$

does exactly this.

The choice $E = c_{00} \subset \ell_1$ makes the coordinate functionals π_n vanish eventually on every fixed vector. More importantly, they vanish uniformly on compact subsets of c_{00}. Hence every continuous $f : \beta\mathbb{N} \to c_{00}$ has $\pi_n(f(n)) \to 0$, because its range is compact. This gives a continuous diagonal map from $C_p(\beta\mathbb{N}, c_{00})$ into $(c_0)_p$.

In the other direction, a null sequence $a = (a_n) \in c_0$ gives a continuous map on $\beta\mathbb{N}$ by sending n to $a_n e_n$ and the entire remainder $\beta\mathbb{N} \setminus \mathbb{N}$ to zero. These two maps compose to the identity on $(c_0)_p$. Thus the vector-valued function space contains a complemented $(c_0)_p$, although both factors separately fail the JNP.

4 The counterexample

Lemma 1 (uniform coordinate decay on compacta)**.** *If $K \subset c_{00}$ is compact for the inherited ℓ_1 norm, then*

$$\sup_{x\in K} |\pi_n(x)| \longrightarrow 0.$$

Proof. For every fixed $x \in c_{00}$, one has $\pi_n(x) = 0$ for all sufficiently large n, and $\|\pi_n\| \le 1$ on $c_{00} \subset \ell_1$. Given $\varepsilon > 0$, choose a finite $\varepsilon/2$-net $x^1, \dots, x^m$ for K. For all sufficiently large n, every $\pi_n(x^j)$ is zero. For such n and any $x \in K$, choose j with $\|x - x^j\|_1 < \varepsilon/2$ to get $|\pi_n(x)| < \varepsilon/2$. □

Lemma 2 (c_{00} is b-feral)**.** *Every barrel-bounded subset of c_{00} is contained in a finite-dimensional subspace. Consequently, c_{00} with the inherited ℓ_1 norm does not have the JNP.*

Proof. Suppose that a set $B \subset c_{00}$ has infinite-dimensional linear span. Then the union of the supports of its elements is infinite. Choose distinct indices n_k and vectors $x^k \in B$ such that $x^k_{n_k} \neq 0$. Choose positive weights w_n satisfying

$$w_{n_k}|x^k_{n_k}| > k$$

and, for instance, put $w_n = 1$ at the remaining indices. The formula

$$q(x) = \sum_n w_n |x_n| \qquad (x \in c_{00})$$

is finite for every x. Its unit ball $U = \{x : q(x) \le 1\}$ is absolutely convex and absorbing. It is also closed in the inherited ℓ_1 topology: q is the supremum of its continuous finite partial sums, hence is lower semicontinuous. Thus U is a barrel. But $q(x^k) > k$, so no scalar multiple of U contains B. Therefore B is not barrel-bounded. This proves the first assertion.

Now let $(\lambda_j) \subset (c_{00})'$ be weak-star null and let B be barrel-bounded. The preceding paragraph places B in a finite-dimensional subspace F. Since the ordinary ℓ_1 unit ball is a barrel, B is norm-bounded. Pointwise convergence of $\lambda_j|_F$ to zero is uniform on norm-bounded subsets of the finite-dimensional space F. Hence $\sup_{x\in B} |\lambda_j(x)| \to 0$. Thus every weak-star null sequence is β^*-null, and c_{00} has no JNP. □

Proposition 1. *$C_p(\beta\mathbb{N})$ does not have the JNP.*

Proof. This fact is recorded explicitly by Banakh and Gabriyelyan [2, Introduction], referring to the normalized finitely supported measure characterization of the JNP for C_p-spaces. Here is a short independent reduction to classical results. If $C_p(\beta\mathbb{N})$ had the JNP, that characterization would give finitely supported measures $\mu_j \in C(\beta\mathbb{N})'$ such that

$$\|\mu_j\| = 1, \qquad \mu_j(f) \longrightarrow 0 \quad (f \in C(\beta\mathbb{N})).$$

The isometry $C(\beta\mathbb{N}) = \ell_\infty$ and Grothendieck's theorem for ℓ_∞ [3] imply that (μ_j) is weakly null in $C(\beta\mathbb{N})'$. The union S of the finite supports of the μ_j is countable, and the closed linear span of the point masses $\{\delta_s : s \in S\}$ is isometric to $\ell_1(S)$. By Hahn–Banach, weak nullity in the ambient measure space implies weak nullity in this $\ell_1(S)$. Schur's theorem [4] then gives $\|\mu_j\| \to 0$, a contradiction. $\square$

Theorem 1 (negative answer to Problem 4.8)**.** *Let*

$$X = \beta\mathbb{N}, \qquad E = (c_{00}, \|\cdot\|_1), \qquad H = C_p(\beta\mathbb{N}, E).$$

Then neither $C_p(X)$ nor E has the Josefson–Nissenzweig property, while H has the JNP. In fact, H contains a complemented subspace linearly homeomorphic to $(c_0)_p$.

Proof. The two negative factor assertions are Lemma 2 and Proposition 1. It remains to construct the complemented copy and a JNP witness.

For $a = (a_n) \in c_0$, define $S(a) : \beta\mathbb{N} \to c_{00}$ by

$$S(a)(n) = a_n e_n \quad (n \in \mathbb{N}), \qquad S(a)(p) = 0 \quad (p \in \beta\mathbb{N} \setminus \mathbb{N}). \tag{1}$$

This map is continuous in the norm topology of c_{00}. Continuity at the isolated points of $\mathbb{N}$ is immediate. If $p \in \beta\mathbb{N} \setminus \mathbb{N}$ and $\varepsilon > 0$, the set $F = \{n : |a_n| \geq \varepsilon\}$ is finite. The clopen neighborhood $\beta\mathbb{N} \setminus F$ of p is mapped into the open ε-ball about zero. Hence $S(a) \in H$.

The linear map

$$S : (c_0)_p \longrightarrow H$$

is continuous for the pointwise topologies: evaluation at $n \in \mathbb{N}$ is the coordinate map $a \mapsto a_n e_n$, while evaluation at every remainder point is zero.

Define

$$D : H \longrightarrow \mathbb{K}^{\mathbb{N}}, \qquad D(f) = (\pi_n(f(n)))_{n\in\mathbb{N}}. \tag{2}$$

For $f \in H$, the range $K = f(\beta\mathbb{N})$ is a norm-compact subset of c_{00}. Lemma 1 yields

$$|\pi_n(f(n))| \leq \sup_{x\in K} |\pi_n(x)| \longrightarrow 0.$$

Thus $D(f) \in c_0$. The map $D : H \to (c_0)_p$ is continuous because its nth coordinate is the continuous functional $\pi_n \circ \mathrm{ev}_n$ on H. Equations (1) and (2) give

$$D \circ S = \mathrm{id}_{(c_0)_p}.$$

Consequently, S is a linear homeomorphism onto its range and $P = S \circ D$ is a continuous projection of H onto $S((c_0)_p)$.

For completeness, the same maps give a direct JNP witness. Put

$$\Phi_n = \pi_n \circ \mathrm{ev}_n \in H'.$$

For each $f \in H$, the preceding compact-range argument says $\Phi_n(f) \to 0$, so (Φ_n) is weak-star null. The set $A = \{e_n : n \in \mathbb{N}\}$ is barrel-bounded in $(c_0)_p$: every barrel in $(c_0)_p$ is also a closed, absolutely convex, absorbing subset of the Banach space c_0, hence is a norm neighborhood, and the set A is norm-bounded. Continuous linear maps carry barrel-bounded sets to barrel-bounded sets, so $S(A)$ is barrel-bounded in H. Finally,

$$\Phi_n(S(e_n)) = 1$$

for every n. Hence (Φ_n) is not $\beta^*(H', H)$-null, proving that H has the JNP. $\square$

5 Scope, novelty, and verification

The example disproves the converse exactly as stated. It exploits the incompleteness of c_{00} in an essential visible way: the completion ℓ_1 has the JNP by the classical Josefson–Nissenzweig theorem. The argument therefore does not decide whether a positive converse may hold when E is complete, barrelled, Frechet, or Banach. Nor does it classify all mixed JNP witnesses in $L_p(X) \otimes E'$.

The proof is noncomputational. The main verifier checkpoints are:

(1) $S(a)$ is norm-continuous at every point of the Stone–Cech remainder;

(2) compact subsets of $c_{00} \subset \ell_1$ have uniform coordinate decay;

(3) D lands in c_0 and is continuous for the pointwise topology;

(4) the weighted lower-semicontinuous gauge really gives a barrel proving that c_{00} is b-feral;

(5) the unit vectors are barrel-bounded in $(c_0)_p$.

Each point is proved explicitly above.

A bounded novelty search was performed on 19 July 2026. It covered the run's registry, solution, attempt, and proof-gap indexes; arXiv/web searches for the exact source title and Problem 4.8; the phrases "$C_p(X, E)$ JNP converse", "$C_p(\beta\omega, c_{00})$", "neither factor has the JNP", and close ASCII variants; arXiv ids 2107.03211 and 2003.06764; and the published 2024 article page. The published source still states that the converse is unknown. The search found the known ingredients that $C_p(\beta\mathbb{N})$ and $c_{00} \subset \ell_1$ fail the JNP, but no paper assembling the diagonal construction above or claiming an answer to Problem 4.8. Novelty is therefore plausible, not certified.

Human-review recommendation. Send to a specialist for review as a likely valid full counterexample. The highest-priority check is the interaction between compact ranges in the incomplete normed space c_{00} and the diagonal map D; Lemma 1 isolates the entire issue.

References

[1] C. Bargetz, J. Kąkol, and D. Sobota, *On complemented copies of the space c_0 in spaces $C_p(X, E)$*, arXiv:2107.03211v3; Math. Nachr. 297 (2024), 644–656.

[2] T. Banakh and S. Gabriyelyan, *The Josefson–Nissenzweig property for locally convex spaces*, arXiv:2003.06764; Filomat 37 (2023), no. 8, 2517–2529.

[3] A. Grothendieck, *Sur les applications linéaires faiblement compactes d'espaces du type $C(K)$*, Canad. J. Math. 5 (1953), 129–173.

[4] I. Schur, *Über lineare Transformationen in der Theorie der unendlichen Reihen*, J. Reine Angew. Math. 151 (1921), 79–111.